Probabilistic Agency: A Pragmatist Epistemology

Epistemology asks what separates opinion from justified belief, what it means to know something rather than merely assert it. This note argues for pragmatism as the epistemological foundation best suited to a probabilistic, adaptive view of reality. It is part of a series. The companion notes on randomness and coherent signaling should be read first.

1. Epistemologies Compared

The major epistemological traditions each define justified belief differently. The table below sketches the landscape, not to dismiss any tradition, but to establish a reference point for what follows.

School	Knowledge is...	Truth is...	Core weakness
Rationalism	Justified belief via reason alone	Logical coherence or self-evidence	Overtrusts abstraction; ignores lived reality
Empiricism	Sense data processed by the mind	What corresponds to observation	Undervalues theory; struggles with abstraction
Skepticism	Uncertain, provisional at best	Potentially unreachable	Risks paralysis or nihilism
Idealism	Mental constructs shaped by mind	What fits a coherent worldview	Risks solipsism; disconnects from feedback
Realism	Objective facts independent of mind	Whatever exists 'out there'	Ignores observer limitations and framing
Pragmatism	Adaptive tools for action, subject to revision	What works reliably in context and survives feedback	Can seem vague; resists universal criteria

Each tradition has its domain of validity and its characteristic blind spot. What matters here is not which is correct in the abstract. That question may be unanswerable. What is most useful for reasoning about complex, adaptive systems is pragmatism, for reasons the rest of this note develops.

2. Facts, Beliefs, and the Limits of Correspondence

A common assumption in the modern world, likely an artifact of the scientific revolution, is that facts define justified beliefs. Philosophy pushes back on this. Facts do not justify beliefs in and of themselves. They are inputs to a reasoning process that is always shaped by context, values, and prior models.

Consider water scarcity. It is a fact that water is a finite resource. That fact alone tells you almost nothing about how water should be allocated. Beliefs about allocation depend on relative abundance in a specific location, the nature of competing uses, the time horizon under consideration, and a dozen other contextual factors. The fact is stable. The justified belief built on it is not. It shifts as context shifts. This is not a weakness in the reasoning. It is an accurate reflection of how reality works.

This is why epistemology cannot be collapsed into methodology. Knowing how to gather facts is not the same as knowing what the facts mean or which facts matter. Those questions require an epistemological framework, and the choice of framework is consequential.

3. Pragmatism: Truth as Convergent Approximation

Pragmatism is sometimes conflated with mere practicality, with doing whatever works in the short run. This misreads it. Pragmatism is a rigorous epistemological position with a specific claim: that truth is not a fixed correspondence between proposition and world, but a dynamic property of beliefs that survive sustained engagement with reality.

William James put it directly: “The truth is what works, in the long run.” C.S. Peirce framed it structurally: truth is the end of inquiry, not static but convergent. Together these statements identify what makes pragmatism distinctive. It refuses final formulation. It builds in revision as a feature, not a failure. There are no non-trivial absolute truths, only approximations that hold until the world pushes back hard enough to require updating.

In this sense pragmatism is structurally similar to Bayesian reasoning, evolutionary adaptation, and engineering safety margins. All three treat current best models as provisional, subject to revision on new evidence. All three build error tolerance into the system rather than assuming it away. The parallel is not accidental. Each is a response to the same underlying reality, that the world is probabilistic and that any system ignoring this will eventually be corrected by it.

4. Language, Models, and the Approximation Problem

Pragmatism addresses a weakness that afflicts all knowledge systems, including mathematics and formal logic: the limits of the language in which they are expressed. As Wittgenstein observed, and as the companion note on coherent signaling develops, the limits of a language are the limits of the world it can describe. We do not have direct access to reality. We have access to what our senses, extended by instrumentation, and our cognitive systems, local and distributed, allow us to perceive. Our models of reality are always approximations. They will always have errors and blind spots.

What science has done, with remarkable success, is make those approximations precise and reliable through reductive mechanism. In a sense, science makes reality simple enough to reason about. But simplification is always lossy. The map is not the terrain, and the map that is most useful for navigation is not necessarily the one that is most faithful to every feature of the landscape.

W.V.O. Quine pushed this further: no belief, not even logic or mathematics, is immune to revision. That insight unsettled analytic philosophy, but adaptive systems have always operated this way. Logic is a powerful tool, but it remains contextual and environmentally dependent. The world pushes back, and only feedback decides what holds.

5. Knowledge as Adaptive Information

From a pragmatist perspective, knowledge is not a static body of verified propositions. It is information and cognition that remain adaptive, capable of updating in response to feedback, resistant to brittle certainty, oriented toward function rather than correspondence.

This maps directly onto the architecture described in the companion notes. Randomness, structured potential operating within constraints, is not the enemy of knowledge but its precondition. Coherent signaling across systems is not achieved by eliminating uncertainty but by operating productively within it. A pragmatist epistemology is the philosophical name for what slime molds, bacteria, songbirds, and scientists do when they are working well: sample the environment, update the model, act on the best current approximation, and remain open to revision.

Entropy sets the stage. Agency makes the move. Knowledge is what accumulates in between, provisional, functional, and always answerable to the world.

This is just one of a series of notes. Refer to the master index for an overview.